**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 246 · Computational problem recognition

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. Why may Spiral be preferable for an uncertain high-risk project?

Q2. What is abstraction?

Q3. What is a base case?

Q4. For C# array int[] x={2,4,6,8}, a loop doubles elements at even indexes (0,2). Give the final array and sum.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

problem recognition — Determine what problem is being solved and its computational form.

decomposition — Split into manageable subproblems.

divide and conquer — Divide input, solve smaller instances, combine results.

constraint — Condition a valid solution must satisfy.

tractability — Whether an algorithm can solve realistic sizes with feasible resources.

## New learning

Recognise ambiguity — 'Fair' is subjective until objective criteria are supplied, such as balancing first-period lessons across teachers.

Make requirements testable — Represent rooms, lessons and slots; specify hard constraints and a success test.

Select techniques appropriately — Decompose into assign/check/evaluate modules. Divide and conquer is useful where subproblems can be solved and combined consistently; not every decomposition is divide and conquer.

## Worked example

Rewrite 'choose the best delivery van route' into a computational task.

1. Predict the method — Replace vague 'best' with a measurable objective such as minimum total travel time.

2. Check and explain — Specify route constraints and verify a produced route covers all required locations.

# We Do · Practise with guidance

A school wants the 'fairest' allocation of laptops across classes. Make it computable.

```
Inputs: stock, demand per class, priorities
Output: allocations respecting stock
Criterion: minimise unmet essential demand
```

Predict the method

Use explicit demand and stock data and define what fairness means with measurable priorities.

Check and explain

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Check total allocated never exceeds stock and each class allocation follows the stated policy.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

For a system allocating parking spaces, give one input, one constraint, one output and one test of correctness.

### Question 2

Explain why 'select the nicest library books' is not computable as stated; propose precise inputs, a ranking rule and an output.

### Question 3

Compare decomposition and divide-and-conquer for sorting 1,000 items, using merge sort as an example.

# Exit Ticket · Answer before checking

An unclear human judgement becomes computable simply by choosing a faster processor.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. It prioritises identifying and addressing risks in each cycle.

Q2. Selecting relevant detail and omitting irrelevant detail.

Q3. A condition that ends recursion.

Q4. Final [4,4,12,8]. Sum=28. Indexes 0 and 2 changed; 1 and 3 unchanged.

### You Do pages · Independent questions

Q1. [4] Input available bays/vehicle requests [1]; constraint one vehicle per bay [1]; output bay assignments [1]; test no duplicate assignments [1].

Q2. [6] 'Nicest' is subjective [1]; need measurable criteria [1], such as age-suitable genre [1] and prior loan count [1]; rank by agreed score [1]; output ordered ISBN list [1].

Q3. [6] Decomposition splits the overall task into subproblems [1]; divide-and-conquer recursively splits input [1], sorts sublists [1] and merges them [1]; combined result is sorted [1], recurrence gives O(n log n) work for merge sort [1].

### Exit Ticket · Model answer

False. It must first be expressed with measurable inputs, rules and success criteria.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

**Do Now** Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding